Provable Benefits of Score Matching

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Based on:

Provable Benefits of Score Matching
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Statistical Efficiency of Score Matching: The View from Isoperimetry
A. Block, Y. Chen, A. Risteski

Outline

Motivation

Energy-based model and partition function

Energy-based models (EBMs) can be parametrized by a class of energies $E_\theta(x)$ written as

$$p_\theta(x) = \frac{\exp(-E_\theta(x))}{Z_\theta},$$

up to a constant of proportionality Z_θ that is called the partition function.

- EBMs are expressive and natural;
- Major challenge: designing efficient algorithms for fitting the models to data.

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- For EBMs, both evaluating the log-likelihood and computing its gradient with respect to θ seem to require computing the partition function Z_θ.
- Specifically, need the gradient $\nabla_\theta \log Z_\theta = -\mathbb{E}_{p_\theta}[\nabla_\theta E_\theta(x)]$;
- Estimation using Markov chains can require many steps to mix in high-dimensional settings.

Score matching

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Given samples from p, the score matching objective is

$$L_{SM}(\theta) = \frac{1}{2}\widehat{\mathbb{E}}_{x \sim p}\left[\|\nabla \log p(x) - \nabla \log p_\theta(x)\|^2\right] + K_p.$$

By integration by parts,

$$L_{SM}(\theta) = \widehat{\mathbb{E}}_{x \sim p}\left[\operatorname{Tr} \nabla^2 \log p_\theta(x) + \frac{1}{2}\|\nabla \log p_\theta(x)\|^2\right],$$

where K_p depends only on p.

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where K_p depends only on p.

Key point: $\nabla_x \log p_\theta(x)$ does not depend on Z_θ.

When does score matching help?

Statistically, by Koehler, Heckett, and Risteski 2022

- Score matching can be statistically comparable to MLE when the target distribution has good geometry (small isoperimetric / Poincaré / log-Sobolev constants).
- However, with bad isoperimetry or bottlenecks, score matching may be statistically worse than MLE.

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In Pabbaraju et al. 2023,

- Construct a natural exponential family where:
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  1. approximating $\log Z_\theta$ is hard,
  2. computing $\nabla_\theta \log Z_\theta$ is hard,
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  unless

$$RP = NP.$$

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- Main insight:

  Score matching can achieve strong statistical guarantees while bypassing the computational burden of the partition function.

Settings

Natural Exponential Family

The paper studies the exponential family

$$p_\theta(x) := \frac{h(x)\exp(\langle \theta, T(x)\rangle)}{Z_\theta},$$

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- The sufficient statistics $T(x) \in \mathbb{R}^{M-1}$ consists of all monomials in $x_1, \ldots, x_n$ of degree at least 1 and at most d ($M = \binom{n+d}{d}$);

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- The base measure $h(x) = \exp\left(-\sum_{i=1}^n x_i^{d+1}\right)$;
- The parameters θ lies in an ℓ_∞-ball: $\theta \in \Theta_B = \{\theta \in \mathbb{R}^{M-1} : \|\theta\|_\infty \leq B\}$.

Why natural exponential family?

- They are classical and widely used models with clear parametrization.
- The gradient of the log-partition function satisfies

$$\nabla_\theta \log Z_\theta = \mathbb{E}_{p_\theta}[T(X)].$$

- For exponential families, the score matching objective becomes a tractable quadratic function of θ.
- This model class allows a rigorous comparison between statistical efficiency and computational tractability.

Statistical efficiency of MLE for exponential family

Given i.i.d. samples $x_1, \ldots, x_n \sim p_\theta$, the maximum likelihood estimator is

$$\hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta' \in \Theta} \widehat{\mathbb{E}}[\log p_{\theta'}(X)]$$

where $\widehat{\mathbb{E}}$ denotes the expectation over the samples.

Theorem 1 (MLE (Van der Vaart 2000))

As $n \to \infty$ and under appropriate regularity conditions,

$$\sqrt{n}\left(\hat{\theta}_{\mathrm{MLE}} - \theta\right) \to N\left(0, \Gamma_{\mathrm{MLE}}\right),$$

where $\Gamma_{\mathrm{MLE}} := \mathcal{I}(\theta^\star)^{-1}$, and $\mathcal{I}(\theta^\star)$ is known as the Fisher information matrix.

For any $\theta^\star \in \mathbb{R}^{M-1}$, the Fisher information matrix of p_θ with respect to the sufficient statistics $T(x)$ is defined as

$$\mathcal{I}(\theta) := \mathbb{E}_{x \sim p_\theta}[T(x)T(x)^\top] - \mathbb{E}_{x \sim p_\theta}[T(x)]\mathbb{E}_{x \sim p_\theta}[T(x)]^\top.$$

Statistical efficiency of score matching for exponential family

For exponential family, the score matching loss is quadratic and

$$\arg\min_{\theta} L_{\mathrm{SM}}(\theta) = -\widehat{\mathbb{E}}_{x\sim p}\left[(JT)_x(JT)_x^\top\right]^{-1}\widehat{\mathbb{E}}_{x\sim p}\Delta T(x),$$

where $(JT)_x \in \mathbb{R}^{(M-1)\times n}$ is the Jacobian of T at x and where $\Delta f = \sum_i \partial_i^2 f$.

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Definition 2 (Restricted Poincaré for exponential families)

The restricted Poincaré constant of $p \in \mathcal{P}_{n,d,B}$ is the smallest $C_P > 0$ such that for all $w \in \mathbb{R}^{M-1}$,

$$\mathrm{Var}_p(\langle w, T(x)\rangle) \le C_P \mathbb{E}_{x\sim p}\left\|\nabla_x \langle w, T(x)\rangle\right\|_2^2 .$$

Statistical efficiency of score matching for exponential family

Theorem 3 (Koehler et al. (2022))

Under suitable regularity conditions, for any p_{θ^*} *with restricted Poincaré constant* C_P *and with* $\lambda_{\min}(\mathcal{I}(\theta^*)) > 0$*, given* N *independent samples* $x^{(1)}, \ldots, x^{(N)} \sim p_{\theta^*}$*, the score matching estimator* $\hat{\theta}_{\mathrm{SM}} = \hat{\theta}_{\mathrm{SM}}(x^{(1)}, \ldots, x^{(N)})$ *satisfies*

$$\sqrt{N}\left(\hat{\theta}_{\mathrm{SM}} - \theta^*\right) \xrightarrow{d} \mathcal{N}(0, \Gamma),$$

where

$$\|\Gamma\|_{\mathrm{op}} \leq \frac{2C_P^2\left(\|\theta^*\|_2^2 \mathbb{E}_{x \sim p_{\theta^*}}\|(JT)(x)\|_{\mathrm{op}}^4 + \mathbb{E}_{x \sim p_{\theta^*}}\|\Delta T(x)\|_2^2\right)}{\lambda_{\min}(\mathcal{I}(\theta^*))^2}.$$

Moreover,

$$(JT)(x)_i = \nabla_x T_i(x), \qquad \Delta T(x) = \mathrm{Tr}\, \nabla_x^2 T(x).$$

MLE vs Score Matching

Score matching is statistically efficient when

1. The distribution has good geometry;
2. The sufficient statistics are smooth (terms like $\mathbb{E}_{x \sim p_{\theta^*}} \|(JT)(x)\|_{\mathrm{op}}^4$ and $\mathbb{E}_{x \sim p_{\theta^*}} \|\Delta T(x)\|_2^2$ measure smoothness of T);
3. The Fisher information is well-conditioned.

MLE vs Score Matching

Outline of the proof in Koehler, Heckett, and Risteski 2022:

1. **Start from the asymptotic covariance formula.**
   By Forbes and Lauritzen 2015,
   $$\Gamma_{\mathrm{SM}} = \mathbb{E}[(JT)_X(JT)_X^\top]^{-1}\Sigma_{(JT)_X(JT)_X^\top\theta^\star+\Delta T}\mathbb{E}[(JT)_X(JT)_X^\top]^{-1}.$$
2. **Compare the score-matching matrix with Fisher information.**
   By Lemma 1 and the Poincaré inequality,
   $$\mathbb{E}[(JT)_X(JT)_X^\top]^{-1} \preceq C_P\mathcal{I}(\theta^\star)^{-1}.$$
3. **Control the middle covariance term.**
   By Lemma 2, $\Sigma_{A+B} \preceq 2\Sigma_A + 2\Sigma_B$. Apply this with $A = (JT)_X(JT)_X^\top\theta$, $B = \Delta T$.
4. **Use operator-norm bounds.**
   $\|\Sigma_A\|_{\mathrm{op}} \leq \|\theta\|_2^2\mathbb{E}\|(JT)_X\|_{\mathrm{op}}^4$, $\|\Sigma_B\|_{\mathrm{op}} \leq \mathbb{E}\|\Delta T\|_2^2$.
5. **Combine the bounds.** The two inverse matrices contribute two factors of $C_P\mathcal{I}(\theta^\star)^{-1}$, giving the final dependence on $C_P^2\|\mathcal{I}(\theta^\star)^{-1}\|_{\mathrm{op}}^2 = \frac{C_P^2}{\lambda_{\min}(\mathcal{I}(\theta^\star))^2}$.

Hardness of implementing optimization oracles

Background: NP-Hardness

- A problem is in NP (nondeterministic polynomial time) if a proposed solution can be checked in polynomial time.
- A problem is NP-hard if every problem in NP can be reduced to it in polynomial time.
- 3-SAT is a canonical NP-complete problem: it is both in NP and NP-hard.
- To prove hardness, one reduces from 3-SAT:

$$\text{3-SAT} \leq_p \text{Target Problem.}$$

- For randomized algorithms, this would imply

$$RP = NP,$$

where *RP* is *randomized polynomial time* with one-sided error.

In this paper:

The target problems are MLE-related computations: (1) sampling from p_θ, (2) approximating $\log Z_\theta$, and (3) computing $\nabla_\theta \log Z_\theta$.

Definition of the 3-SAT problem

Definition 4 (Clause/formula polynomials)

Given a 3-clause formula of the form
$$C = \tilde{x}_i \vee \tilde{x}_j \vee \tilde{x}_k,$$
where $\tilde{x}_i = x_i$ or $\tilde{x}_i = -x_i$, we construct a polynomial $H_C \in \mathbb{R}[x_1, \ldots, x_n]_{\leq 6}$ defined by
$$H_C(x) = f_i(x_i)^2 f_j(x_j)^2 f_k(x_k)^2,$$
where
$$f_i(t) = \begin{cases} t+1, & \text{if } x_i \text{ is negated in } C, \\ t-1, & \text{otherwise.} \end{cases}$$
For example, if $C = x_1 \vee x_2 \vee -x_3$, then $H_C = (x_1 - 1)^2(x_2 - 1)^2(x_3 + 1)^2$.
Further, given a 3-SAT formula $C = C_1 \wedge \cdots \wedge C_m$ on m clauses, we define the polynomial
$$H_C(x) = H_{C_1}(x) + \cdots + H_{C_m}(x).$$

It can be seen that any $x \in \mathcal{H} := \{-1, 1\}^n \subseteq \mathbb{R}^n$ corresponds to a satisfying assignment for C if and only if

$$H_C(x) = 0.$$

Note that there are possibly points outside $\mathcal{H}$ which satisfy $H_C(x) = 0$. To avoid these solutions, we introduce another polynomial.

Definition 5 (Hypercube polynomial)

We define $G : \mathbb{R}^n \to \mathbb{R}$ by

$$G(x) = \sum_{i=1}^{n} (1 - x_i^2)^2.$$

For any $\alpha, \beta > 0$, the roots in $\mathbb{R}^n$ of the polynomial

$$F_C(x) = \alpha H_C(x) + \beta G(x)$$

are precisely the vertices of $\mathcal{H}$ that correspond to satisfying assignments for C.

Definition 6

Let C be a 3-CNF formula with n variables and m clauses. Let $\alpha, \beta > 0$. Then we define a distribution $P_{C,\alpha,\beta}$ with density function

$$p_{C,\alpha,\beta}(x) := \frac{h(x)\exp(-\alpha H_C(x) - \beta G(x))}{Z_{C,\alpha,\beta}},$$

where

$$Z_{C,\alpha,\beta} = \int_{\mathbb{R}^n} h(x)\exp(-\alpha H_C(x) - \beta G(x))\, dx.$$

Idea of the Hardness Construction

The paper reduces a discrete 3-SAT problem to a continuous distribution:

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- The constructed density $p_{C,\alpha,\beta}(x) \propto h(x) \exp(-\alpha H_C(x) - \beta G(x))$ lies in the exponential family $\mathcal{P}_{n,d,B}$, for $d = 7$ and $B = \Omega(\beta + m\alpha)$ (Lemma A.2).

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- If $\theta(C, \alpha, \beta)$ is the parameter vector that induces $P_{C,\alpha,\beta}$, it suffices to show that
  - (a) approximating $\log Z_{\theta(C,\alpha,\beta)}$,
  - (b) approximating $\nabla_\theta \log Z_{\theta(C,\alpha,\beta)}$, and
  - (c) sampling from $P_{C,\alpha,\beta}$

  are NP-hard under randomized reductions.

Hardness of implementing optimization oracles

Theorem A.1 (Valiant and Vazirani 1985; Cook 2023)

Suppose that there is a randomized poly(n)-time algorithm for the following problem: given a 3-CNF formula C with n variables and at most $5n$ clauses, under the promise that C has at most one satisfying assignment, determine whether C is satisfiable. Then

$$NP = RP.$$

Hardness of implementing optimization oracles

Given a point $v \in \mathcal{H}$, let $\mathcal{O}(v) := \{x \in \mathbb{R}^n : x_i v_i \geq 0, \ \forall i \in [n]\}$ denote the octant containing v, and let $\mathcal{B}_r(v) := \{x \in \mathbb{R}^n : \|x - v\|_\infty \leq r\}$ denote the ball of radius r with respect to ℓ_∞ norm.

Lemma A.3

Let $p := p_{C,\alpha,\beta}$ and $Z := Z_{C,\alpha,\beta}$ for some 3-CNF C with m clauses and n variables, and some parameters $\alpha, \beta > 0$. Let $r \in (0,1)$. If $\beta \geq 40r^{-2}\log(4n/r)$, then for any $v \in \mathcal{H}$ that is a satisfying assignment for C,

$$\Pr_{x \sim p}\left(x \in \mathcal{B}_r(v)\right) \geq \frac{e^{-1-81m\alpha r^2}}{Z}\left(\int_0^\infty \exp\left(-x^8 - \beta(1-x^2)^2\right)\, dx\right)^n.$$

For any $w \in \mathcal{H}$ that is not a satisfying assignment for C,

$$\Pr_{x \sim p}\left(x \in \mathcal{O}(w)\right) \leq \frac{e^{-\alpha}}{Z}\left(\int_0^\infty \exp\left(-x^8 - \beta(1-x^2)^2\right)\, dx\right)^n.$$

Hardness of approximating $\log Z_{C,\alpha,\beta}$:

We bound the mass of $P_{C,\alpha,\beta}$ in each orthant of $\mathbb{R}^n$.

In particular, for

$$\alpha = \Omega(n) \qquad \text{and} \qquad \beta = \Omega(m \log m),$$

the partition function $Z_{C,\alpha,\beta}$ is exponentially larger when the formula C is satisfiable than when it is not.

Then approximating $Z_{C,\alpha,\beta}$ allows distinguishing a satisfiable formula from an unsatisfiable formula, which is NP-hard. This implies the following theorem.

Theorem 7

Fix $n \in \mathbb{N}$ and let $B \geq Cn^2$ for a sufficiently large constant C. Unless $RP = NP$, there is no $\mathrm{poly}(n)$-time algorithm which takes as input an arbitrary $\theta \in \Theta_B$ and outputs an approximation of $\log Z_\theta$ with additive error less than $n \log 1.16$.

Proof Sketch of Theorem 3.4

Lemma A.6

Fix $n, m \in \mathbb{N}$ and let

$$\alpha \geq 2(n+1) \qquad \text{and} \qquad \beta \geq 6480m \log\left(13n\sqrt{m}\right).$$

There is a constant

$$A = A(n, m, \alpha, \beta)$$

so that the following hold for every 3-CNF formula C with n variables and m clauses:

- If C is unsatisfiable, then
$$Z_{C,\alpha,\beta} \leq A.$$
- If C is satisfiable, then
$$Z_{C,\alpha,\beta} \geq (e/2)^n A.$$

Proof Sketch of Theorem 3.4

- Start from the following NP-hard problem: given two 3-CNF formulas C and C' with n variables and at most $10n$ clauses, where exactly one of them is satisfiable, determine which one is satisfiable.
- Choose
$$\alpha = 2(n+1), \qquad \beta = 64800n \log\left(13n\sqrt{10n}\right).$$
- Suppose there exists a poly(n)-time algorithm that, for any $\theta \in \Theta_B$, approximates $\log Z_\theta$ with additive error less than $n \log 1.16$.
- For the two formulas C and C', construct parameters $\theta = \theta(C, \alpha, \beta), \theta' = \theta(C', \alpha, \beta)$.
- Therefore, the assumed algorithm gives approximations $\widetilde{Z}_\theta, \widetilde{Z}_{\theta'}$ with multiplicative error less than 1.16^n.
- By Lemma A.6, the true partition function is separated by a larger exponential gap:
$$Z_{C,\alpha,\beta} \text{ is larger when } C \text{ is satisfiable.}$$
- Hence,
$$\widetilde{Z}_\theta > \widetilde{Z}_{\theta'} \quad \Longleftrightarrow \quad C \text{ is satisfiable.}$$

Hardness of approximating $\nabla_\theta \log Z_{\theta(\mathcal{C},\alpha,\beta)}$

Theorem 3.5

Fix $n \in \mathbb{N}$ and let $B \geq Cn^2 \log(n)$ for a sufficiently large constant C. Unless $RP = NP$, there is no poly(n)-time algorithm which takes as input an arbitrary $\theta \in \Theta_B$ and outputs an approximation of $\nabla_\theta \log Z_\theta$ with additive error, in ℓ_∞ sense, less than $1/20$.

Hardness of approximating $\nabla_\theta \log Z_{\theta(\mathcal{C},\alpha,\beta)}$

Proof sketch:

- For exponential families, the gradient of the log-partition function is $\nabla_\theta \log Z_\theta = \mathbb{E}_{x\sim p_\theta}[T(x)]$.
- Since $T(x)$ contains the coordinate functions, approximating $\nabla_\theta \log Z_\theta$ also approximates the mean $\mathbb{E}_{x\sim p_\theta}[x]$.
- Reduce from the unique satisfying assignment setting: assume C has exactly one satisfying assignment $v^* \in \mathcal{H}$.
- The constructed distribution $P_{C,\alpha,\beta}$ concentrates mass in the orthant corresponding to v^*.
- For suitable $\alpha = \Theta(n)$ and $\beta = \Omega(mn \log m)$, $\mathbb{E}_{x\sim p_{C,\alpha,\beta}}[v_i^* x_i] \geq 1/20$ for all $i \in [n]$.
- Therefore, an ℓ_∞ approximation error smaller than $1/20$ lets us recover every coordinate of the unique satisfying assignment by taking signs: $v_i^* = \mathrm{sign}(\mathbb{E}[x_i])$.
- Thus an efficient algorithm for approximating $\nabla_\theta \log Z_\theta$ would solve unique 3-SAT in randomized polynomial time.
- By Theorem A.1, this would imply $RP = NP$.

Theorem A.5

Let $B \geq Cn^2$ for a sufficiently large constant C. Unless $RP = NP$, there is no algorithm which takes as input an arbitrary $\theta \in \Theta_B$ and outputs a sample from a distribution Q with $TV(P_\theta, Q) \leq 1/3$ in poly(n) time.

Proof sketch:

- Given a 3-CNF formula C with n variables and at most $5n$ clauses, choose

$$\alpha = 2(n+1), \qquad \beta = 32400n \log\left(13n\sqrt{5n}\right).$$

- Construct the parameter vector $\theta = \theta(C, \alpha, \beta)$.
- By Lemma A.2, for $B \geq Cn^2$ with sufficiently large C, we have $\theta \in \Theta_B$.
- By Lemma A.4, if C is satisfiable, then a true sample from $P_{C,\alpha,\beta}$ lands in a satisfying orthant with probability at least $1/2$:

$$\Pr_{x \sim P_{C,\alpha,\beta}}[\text{sign}(x) \text{ satisfies } C] \geq 1/2.$$

Statistical efficiency of MLE and SM

Statistical efficiency of MLE

For any $\theta \in \Theta_B$, we can lower bound the smallest eigenvalue of the Fisher information matrix $\mathcal{I}(\theta)$.

Theorem 4.1

For any $\theta \in \Theta_B$, it holds that

$$\lambda_{\min}(\mathcal{I}(\theta)) \geq (nB)^{-O(d^3)}.$$

As a corollary, given N samples from p_θ, it holds as $N \to \infty$ that $\sqrt{N}\left(\hat{\theta}_{\mathrm{MLE}} - \theta\right) \to N(0, \Gamma_{\mathrm{MLE}})$, where $\|\Gamma_{\mathrm{MLE}}\|_{\mathrm{op}} \leq (nB)^{O(d^3)}$.

Moreover, for sufficiently large N, with probability at least 0.99,

$$\left\|\hat{\theta}_{\mathrm{MLE}} - \theta\right\|_2^2 \leq \frac{(nB)^{O(d^3)}}{N}.$$

Proof of Theorem 4.1

Goal: prove that the Fisher information matrix is well-conditioned:

$$\lambda_{\min}(\mathcal{I}(\theta)) \geq (nB)^{-O(d^3)}.$$

For any direction w, define the polynomial $f(x) = \langle w, T(x) \rangle$. Then $w^\top \!= I(\theta) w \ \mathrm{Var}_{p_\theta}(f)$. So lower bounding $\lambda_{\min}(\mathcal{I}(\theta))$ reduces to showing that a nonzero low-degree polynomial cannot have too small variance under p_θ.

Lemma 4.2

For $f \in \mathbb{R}[x_1, \ldots, x_n]_{\leq d}$,

$$\|f\|_{\mathrm{mon}}^2 \leq \binom{n+d}{d} (16e)^d \|f\|_{L^2([-1,1]^n)}^2.$$

Meaning: if a polynomial has large monomial coefficients, then it must have nontrivial L^2 mass on $[-1, 1]^n$.

Proof of Theorem 4.1

Lemma 4.3

For any $\theta \in \Theta_B$, define $p := p_\theta$ and $R := 2^{d+8} nBM$. Then for any $f \in \mathbb{R}[x_1, \ldots, x_n]_{\le d}$, there exists $z \in \mathbb{R}^n$ with $\|z\|_\infty \le R$ and some $\epsilon \ge 1/(2(d+1)MR^d(n+B))$ such that

$$\mathrm{Var}_p(f) \ge \frac{1}{2e} \mathrm{Var}_{\tilde{U}}(f),$$

where $\tilde{U}$ is uniform on $\{x \in \mathbb{R}^n : \|x - z\|_\infty \le \epsilon\}$.

Meaning: if f has variance under p, we can compare it to the variance on a small uniform box. On a sufficiently small box, the density of p does not fluctuate too much.

Lemma 4.4

For $f \in \mathbb{R}[x_1, \ldots, x_n]_{\le d}$ with $f(0) = 0$, $\mathrm{Var}_p(f) \ge \frac{1}{2^{2d}(d+1)^{2d}(16e)^{d+1}M^{2d+3}R^{2d^2+2d}(n+B)^{2d}} \|f\|_{\mathrm{mon}}^2$.

Meaning: Lemmas 4.2 and 4.3 imply that variance under p_θ controls the monomial coefficient norm.

Statistical Efficiency of Score Matching

Theorem 5.4

Fix $n, d, B, N \in \mathbb{N}$. Pick any $\theta^* \in \Theta_B$ and let $x^{(1)}, \ldots, x^{(N)} \sim p_{\theta^*}$ be independent samples. Then as $N \to \infty$, the score matching estimator $\hat{\theta}_{\mathrm{SM}} = \hat{\theta}_{\mathrm{SM}}(x^{(1)}, \ldots, x^{(N)})$ satisfies

$$\sqrt{N}(\hat{\theta}_{\mathrm{SM}} - \theta^*) \to \mathcal{N}(0, \Gamma),$$

where $\|\Gamma\|_{\mathrm{op}} \leq (nB)^{O(d^3)}$. As a corollary, for all sufficiently large N, it holds with probability at least 0.99 that

$$\|\hat{\theta}_{\mathrm{SM}} - \theta^*\|_2^2 \leq \frac{(nB)^{O(d^3)}}{N}.$$

Interpretation:

- Score matching achieves the same parametric squared-error rate as MLE: $O(1/N)$.
- The asymptotic covariance is polynomially controlled in n and B.
- Unlike MLE, the estimator avoids computing Z_θ and $\nabla_\theta \log Z_\theta$.

Proof Outline of Theorem 5.4

- Apply **Theorem 2.2** from Koehler et al. (2022), with required regularity conditions verified by **Lemma B.4**, and the Fisher information is nonsingular by **Theorem 4.1**,

$$\|\Gamma\|_{\mathrm{op}} \leq \frac{2C_P^2\left(\|\theta^*\|_2^2\mathbb{E}\|(JT)(x)\|_{\mathrm{op}}^4 + \mathbb{E}\|\Delta T(x)\|_2^2\right)}{\lambda_{\min}(\mathcal{I}(\theta^*))^2}.$$

- Bound the restricted Poincaré constant C_P using **Lemma 5.3**:

$$C_P \leq \left(4 + 4\|\mathbb{E}[T(x)]\|_2^2\right)\frac{\lambda_{\max}(\mathcal{I}(\theta^*))}{\min(1, \lambda_{\min}(\mathcal{I}(\theta^*)))}.$$

- Use **Lemma B.2** to bound $\|\mathbb{E}[T(x)]\|_2$ and $\lambda_{\max}(\mathcal{I}(\theta^*))$, and use **Theorem 4.1** to lower bound $\lambda_{\min}(\mathcal{I}(\theta^*))$.
- Therefore, $C_P \leq (nB)^{O(d^3)}$.
- Bound the remaining derivative terms using **Lemma B.3**: $\mathbb{E}\|(JT)(x)\|_{\mathrm{op}}^4, \mathbb{E}\|\Delta T(x)\|_2^2$.
- Substitute all bounds into the covariance inequality to obtain $\|\Gamma\|_{\mathrm{op}} \leq (nB)^{O(d^3)}$.

Conclusions of Pabbaraju et al. 2023

- The paper gives a concrete exponential family:

$$p_\theta(x) \propto h(x) \exp(\langle \theta, T(x) \rangle),$$

  where $T(x)$ consists of bounded-degree polynomial sufficient statistics.
- In this family, score matching and MLE have comparable statistical efficiency, up to polynomial factors.
- However, MLE requires computationally hard tasks, such as approximating $\log Z_\theta$ and $\nabla_\theta \log Z_\theta$.
- Score matching avoids the partition function and can be computed efficiently through a quadratic objective.
- Therefore, the paper gives a separation:

  score matching is computationally easier than MLE,

  while retaining comparable sample efficiency.

Inefficiency of score matching in the presence of sparse cuts

Inefficiency of score matching in the presence of sparse cuts

Consider estimating a distribution p_θ in an exponential family with isoperimetric constant C_{IS}.

Definition 4 of Koehler, Heckett, and Risteski 2022

The isoperimetric constant $C_{\mathrm{IS}}(q)$ is the smallest constant such that, for every set S,

$$\min\left\{\int_S q(x)\,dx,\ \int_{S^c} q(x)\,dx\right\} \leq C_{\mathrm{IS}}(q) \liminf_{\epsilon\to 0} \frac{\int_{S_\epsilon} q(x)\,dx - \int_S q(x)\,dx}{\epsilon}. \tag{5}$$

where $S_\epsilon = \{x : d(x, S) \leq \epsilon\}$, and $d(x, S)$ denotes the Euclidean distance of x from the set S.

The isoperimetric constant is related to the Poincaré constant by $C_P \leq 4C_{\mathrm{IS}}^2$.
Assuming S is chosen so that $\int_S q(x)\,dx < \frac{1}{2}$, the left-hand side can be interpreted as the volume of S, and the right-hand side as the surface area of S with respect to q.

Inefficiency of score matching in the presence of sparse cuts

Then p_θ can be viewed as a member of an enlarged exponential family with one more $O_{\partial S}(1)$-Lipschitz sufficient statistic, such that score matching has asymptotic relative efficiency

$$\Omega_{\partial S}(C_{\mathrm{IS}})$$

compared to the MLE, where ∂S denotes the boundary of the isoperimetric cut of p_θ and $\Omega_{\partial S}$ indicates a constant depending only on the geometry of the manifold ∂S.

Sparse Cuts Can Make Score Matching Inefficient

Theorem 3 of Koehler, Heckett, and Risteski 2022

There exists an absolute constant $c > 0$ such that the following is true. Suppose that $p_{\theta_1^*}$ is an element of an exponential family with sufficient statistic F_1 and parameterized by elements of Θ_1.
Suppose S is a set with smooth boundary ∂S which has reach $\tau_{\partial S} > 0$. Suppose that 1_S is not an affine function of F_1, so there exists $\delta_1 > 0$ such that

$$\sup_{w_1 : \mathrm{Var}(\langle w_1, F_1 \rangle) = 1} \mathrm{Cov}\left(\langle w_1, F_1 \rangle, \frac{1_S}{\sqrt{\mathrm{Var}(1_S)}} \right)^2 \leq 1 - \delta_1. \tag{11}$$

Suppose that $\gamma > 0$ satisfies $\gamma < \min \left\{ \frac{c^d}{(1+\|\theta_1\|) \sup_{x: d(x, \partial S) \leq \gamma} \|(JF_1)_x\|_{\mathrm{OP}}}, \frac{c\tau_{\partial S}}{d} \right\}$, and is small enough so that

$$0 < \delta := 1 - \left(\sqrt{1 - \delta_1} + 2 \sqrt{\frac{\gamma \int_{x \in \partial S} p(x)\, dx}{\Pr(X \in S)(1 - \Pr(X \in S))}} \right)^2 .$$

Sparse Cuts Can Make Score Matching Inefficient

Theorem 3 (continued)

Define an additional sufficient statistic $F_2 = 1_S * \psi_\gamma$ so that the enlarged exponential family contains distributions of the form

$$p_{(\theta_1,\theta_2)}(x) \propto \exp(\langle \theta_1, F_1(x)\rangle + \theta_2 F_2(x)) .$$

Consider the MLE and score matching estimators in this exponential family with ground truth $p_{(\theta_1^*,0)}$. Then there exists some w so that the relative inefficiency of the score matching estimator compared to the MLE for estimating $\langle w, \theta\rangle$ admits the following lower bound:

$$\frac{\langle w, \Gamma_{\mathrm{SM}} w\rangle}{\langle w, \Gamma_{\mathrm{MLE}} w\rangle} \geq \frac{c'}{\gamma} \frac{\min\{\Pr(X \in S), \Pr(X \notin S)\}}{\int_{x\in\partial S} p(x)\, dx},$$

where $c' := \frac{\delta c^d}{1+\|\Sigma_{F_1}\|_{\mathrm{OP}}}$.

Interpretation: If the boundary mass $\int_{\partial S} p(x)\, dx$ is small relative to the mass of S, then the ratio above is large. Thus sparse cuts / bottlenecks can make score matching much less efficient than MLE.

Thank you!

References I

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